JAIGOPAL GARODIA VIVEKANANDA VIDYALAYA AVADI,CHENNAI-600 054

Physics Investigatory Project 2023-2024

Submitted by :

Name: Pandi durai S

Class: XII A

Roll No: 12113

Subject : Physics

Batch: 2023-2024

CERTIFICATE

This is to certify that this project is submitted by PANDI DURAI to the physics department of JAIGOPAL GARODIA VIVEKANANDA VIDYALAYA was carried out by him under the guidance and supervision of MRS. KOKILA during academic session 2023-2024.

Signature of Principal:

Signature of External Examiner:

Signature of Internal Examiner:

ACKNOWLEDGMENT

I wish to express my deep gratitude and sincere thanks to MRS.SREEJA RAMESH, SENIOR SECONDARY PRINCIPAL OF JAIGOPAL GARODIA VIVEKANANDA VIDYALAYA

for her encouragement and for all the facilities that she provided for this project work. I sincerely appreciate this magnanimity by taking me into her fold for which I shall remain indebted to her. I extend my hearty thanks to MRS. KOKILA, Physics Teacher who guided me to do this project successful completion of this project. I take this opportunity to express my deep sense of gratitude for her valuable guidance, constant encouragement, comments, sympathetic attitude and immense motivation which has sustained my effort at all stages of this project work.

DECLARATION

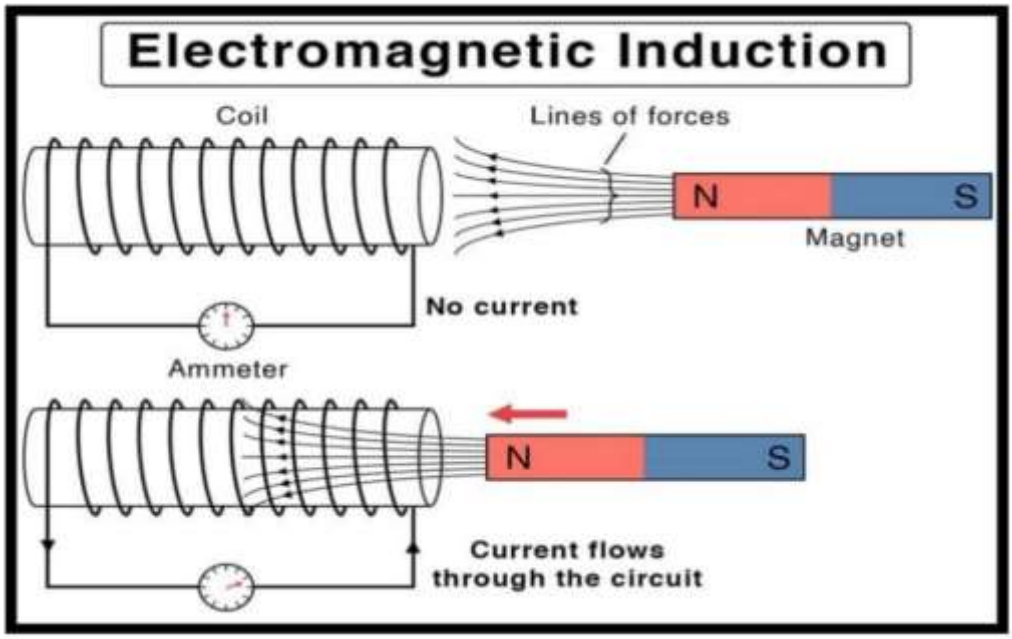
I do hereby declare that this project work has been originally carried under the guidance and supervision of MRS. KOKILA, PHYSICS TEACHER OF JAIGOPAL GARODIA VIVEKANANDA VIDYALAYA, AVADI

S PANDI DURAI

PHYSICS PROJECT

PHENOMENON OF

ELECTROMAGNETIC INDUCTION

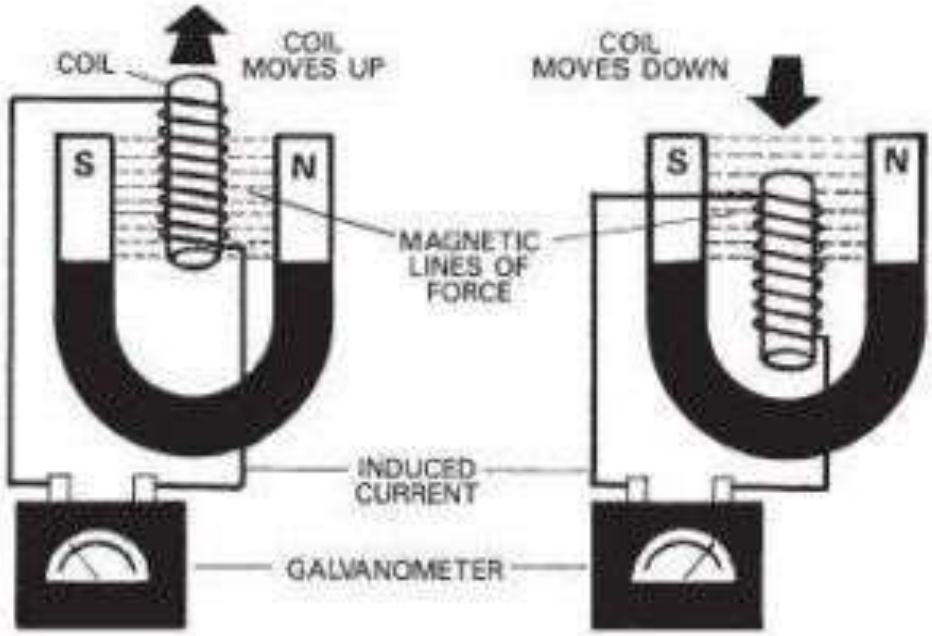


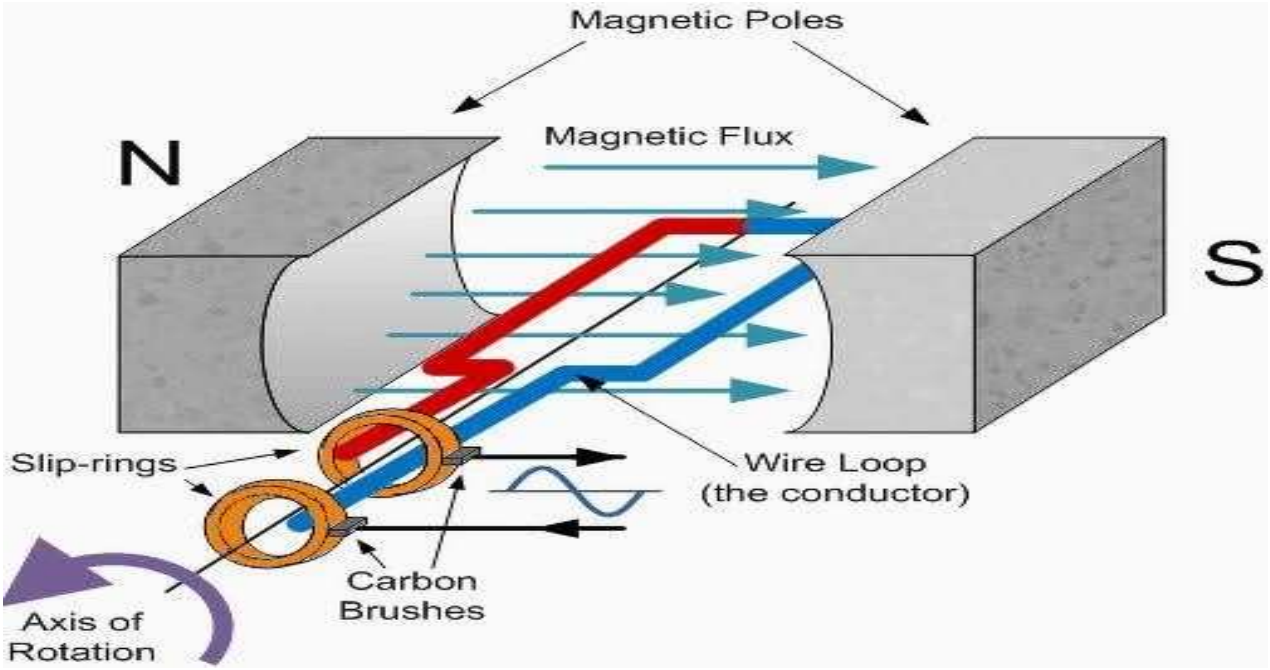
INDEX

- Aim of the Project
- Introduction
- Principle
- Theory
- Applications of EMI
- Observation
- Precautions
- Conclusion
- Experiment
    - Aim
    - Materials Required
    - Procedure
    - Observation
    - Inference
    - Precautions
    - Result
- Bibliography

AIM:

To study the **phenomenon of electromagnetic induction.**





INTRODUCTION:

Electro Magnet:

An electromagnet is a type of magnet in which the magnetic field is produced by electric current. The magnetic field disappears when the current is turned off.

Induction:

This process of generating current in a conductor by placing the conductor in a changing magnetic field is called induction.

Electromagnetic Induction:

Electromagnetic induction is the production of a potential difference (voltage) across a conductor when it is exposed to a varying magnetic field.

Electromagnetic induction is when an electromagnetic field causes molecules in another object to flow. Induction can produce electricity (in coils), heat (in ferrous metals), or waves (in a radio transmitter).

Magnetic Flux:

Magnetic flux is the total number of magnetic field lines passing through a given closed surface. It is defined as the product of the average magnetic field times the perpendicular area it penetrates. It is a scalar quantity and its S.I. unit is Weber (Wb).

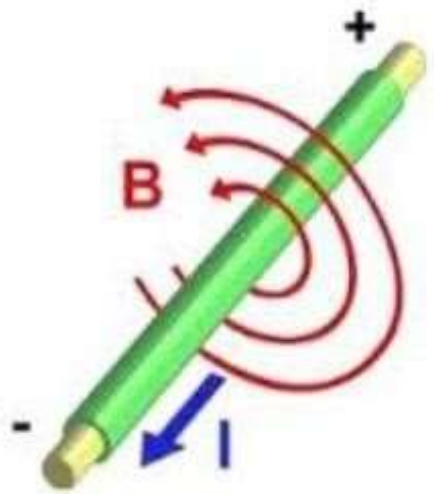
Magnetic flux formula is given by:

$$\Phi = B\,A\cos(\theta)$$

PRINCIPLE:

Electromagnetic induction (or sometime just induction) is a process where a conductor placed in a changing magnetic field (or a conductor moving through a stationary magnetic field) causes the production of a voltage across the conductor. This process of electromagnetic induction, in turn, causes an electrical current - it is said to induce the current.

Electromagnetic (EM) methods use transmitters to create strong time-varying primary magnetic fields that induce electrical currents to flow in conductive rocks



THEORY:

Invention:

Michael Faraday is generally credited with the discovery of induction in 1831 though it may have been anticipated by the work of Francesco Zantedeschi in 1829. Around 1830 to 1832, Joseph Henry made a similar discovery, but did not publish his findings until later

Induced emf:

If magnetic flux through a coil is altered then an emf will be generated in the coil. This effect was first observed and explained by Ampere and Faraday between 1825 and 1831. Faraday discovered that an emf could be generated either by,

a) Moving the coil or the source of flux relative to each other or by
b) Changing the magnitude of the source of magnetic flux in some way.

(Note that the emf is only produced while the flux is changing).

For example, consider two coils as shown in the Figure

1.

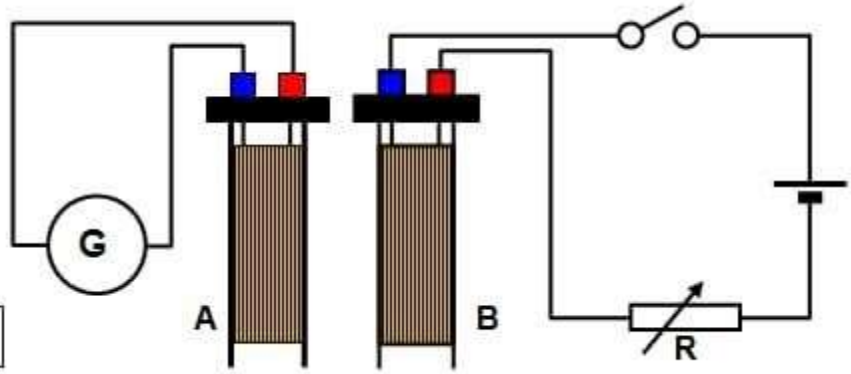


Figure 1

Coil A is connected to a galvanometer and coil B is connected to a battery and has direct current flowing through it. Coil A is within the magnetic field produced by B and an emf can be produced in A by moving the coils relative to each other or by changing the size of the current in B. This can be done by using the rheostat R, switching the current on or off, or (c) using an ac supply for B.

(An emf could also be produced in coil A by replacing coil B with a permanent magnet and moving this relative to coil A.)

REPRESENTATION:

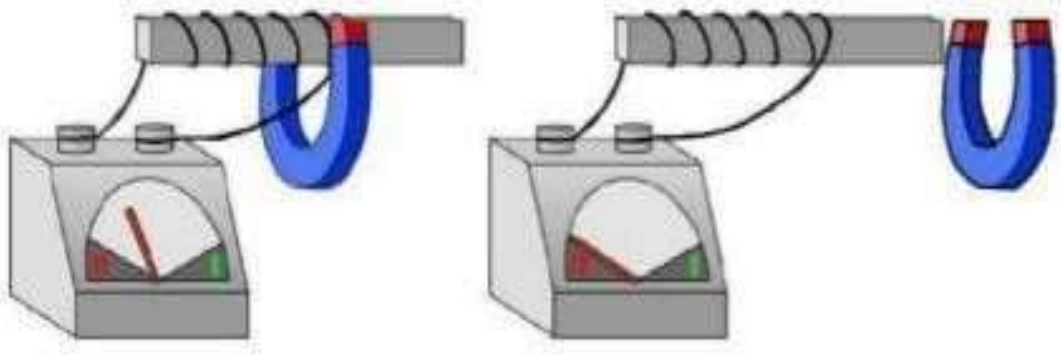
Electromagnetic induction is the production of a potential difference (voltage) across a conductor when it is exposed to a varying magnetic field.

WORKING AND CONSTRUCTION:

Current is produced in a conductor when it is moved through a magnetic field because the magnetic lines of force are applying a force on the free electrons in the conductor and causing them to move. This process of generating current in a conductor by placing the conductor in a changing magnetic field is called induction. This is called induction because there is no physical connection between the conductor and the magnet. The current is said to be induced in the conductor by the magnetic field.

One requirement for this electromagnetic induction to take place is that the conductor, which is often a piece of wire, must be perpendicular to the magnetic lines of force in order to produce the maximum force on the free electrons. The direction that the induced current flows is determined by the direction of the lines of force and by the direction the wire is moving in the field. In the animation above the

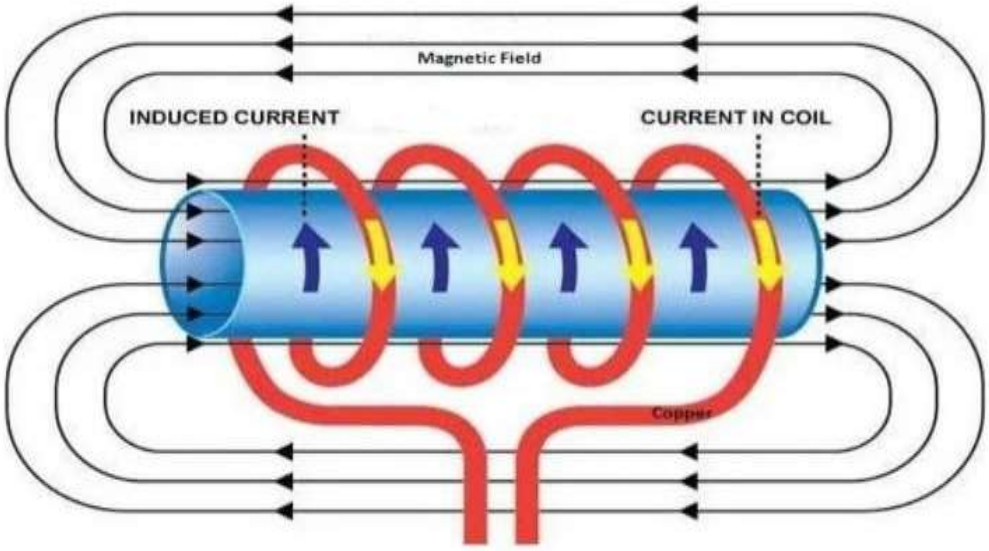
ammeter (the instrument used to measure current) indicates when there is current in the conductor.



If an AC current is fed through a piece of wire, the electromagnetic field that is produced is constantly growing and shrinking due to the constantly changing current in the wire. This growing and shrinking magnetic field can induce electrical current in another wire that is held close to the first wire. The current in the second wire will also be AC and in fact will look very similar to the current flowing in the first wire.

It is common to wrap the wire into a coil to concentrate the strength of the magnetic field at the ends of the coil.

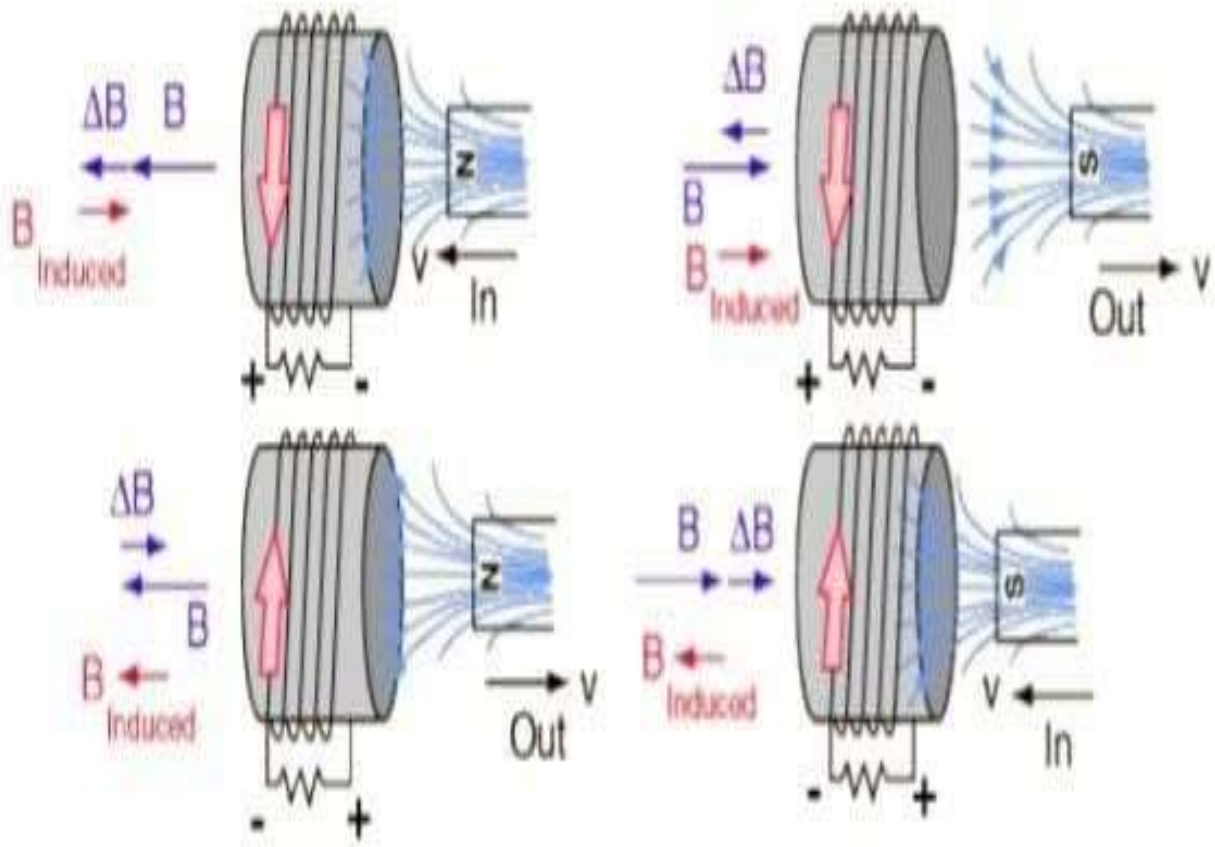
Wrapping the coil around an iron bar will further concentrate the magnetic field in the iron bar. The magnetic field will be strongest inside the bar and at its ends (poles).



Lenz's Law:

When an emf is generated by a change in magnetic flux according to Faraday's Law, the polarity of the induced emf is such that it produces a current whose magnetic field opposes the change which produces it. The induced magnetic field inside any loop of wire always acts to keep the magnetic flux in the loop constant. In the examples below, if the B field is increasing, the induced field acts in opposition to it.

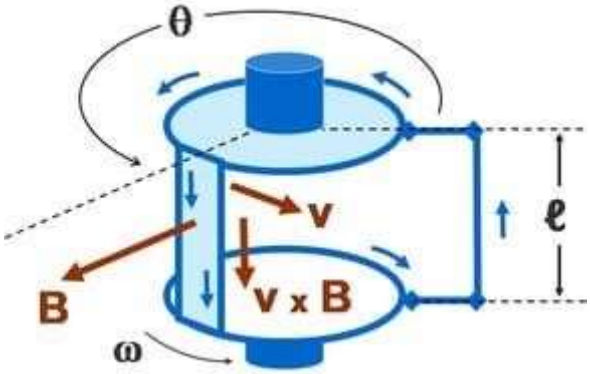
If it is decreasing, the induced field acts in the direction of the applied field to try to keep it constant.



APPLICATIONS OF ELECTROMAGNETIC INDUCTION:

Electrical Generator:

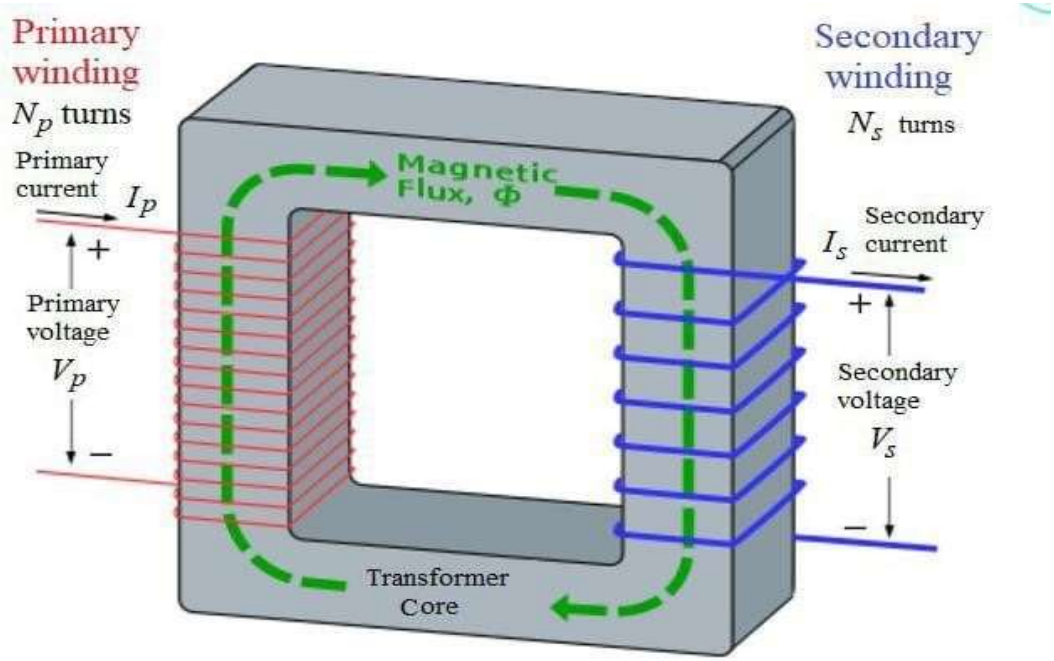
The EMF generated by Faraday's law of induction due to relative movement of a circuit and a magnetic field is the phenomenon underlying electrical generators. When a permanent magnet is moved relative to a conductor, or vice versa, an electromotive force is created. If the wire is connected through an electrical load, current will flow, and thus electrical energy is generated, converting the mechanical energy of motion to electrical energy.



Electrical transformer:

The EMF predicted by Faraday's law is also responsible for electrical transformers. When the electric current in a loop of wire changes, the changing current creates a changing magnetic field. A second wire in reach of this magnetic field will experience this change in magnetic

field as a change in its coupled magnetic flux, d Φ_B / d t. Therefore, an electromotive force is set up in the second loop called the induced EMF or transformer EMF. If the two ends of this loop are connected through an electrical load, current will flow.



Magnetic flow meter:

Faraday's law is used for measuring the flow of electrically conductive liquids and slurries. Such instruments are called magnetic flow meters. The induced voltage *ε generated* in the magnetic field *B* due to a conductive liquid moving at velocity *v* is thus given by:

$$\varepsilon = - Blv$$

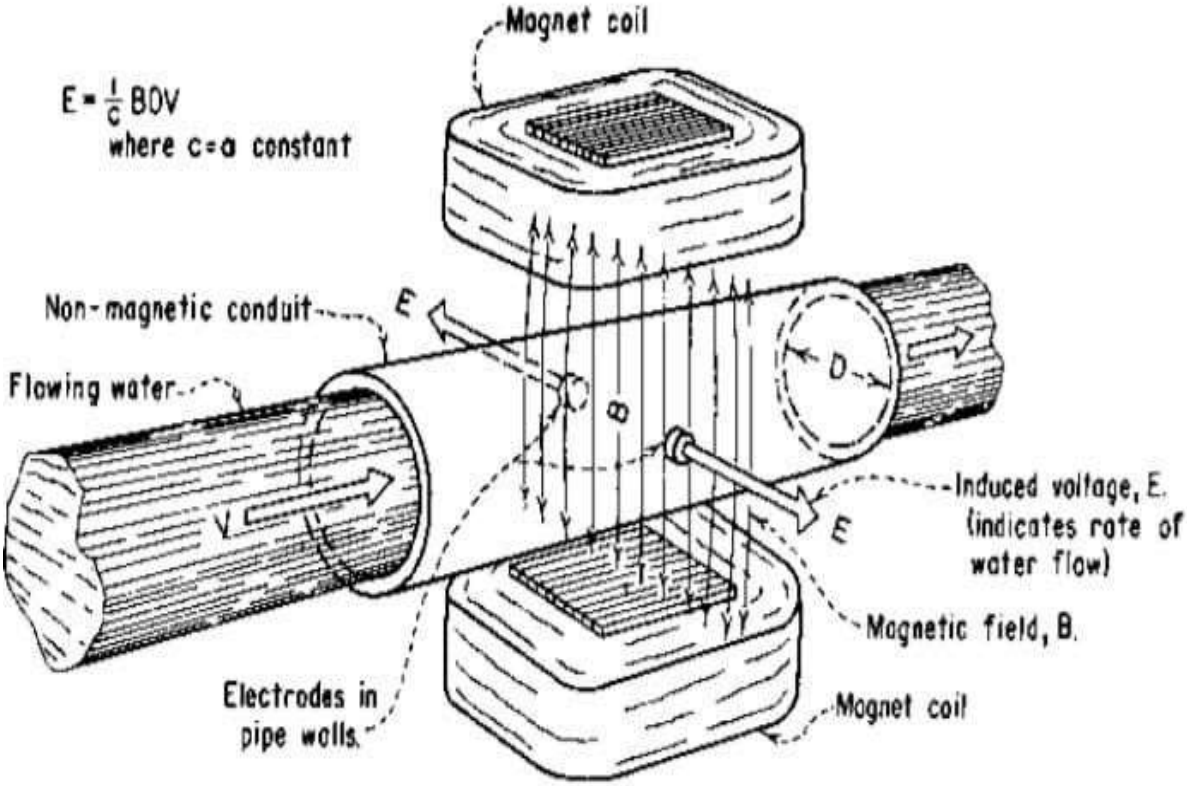
Where,

ε = Motional EMF (volts)

B = Magnetic flux density (Tesla)

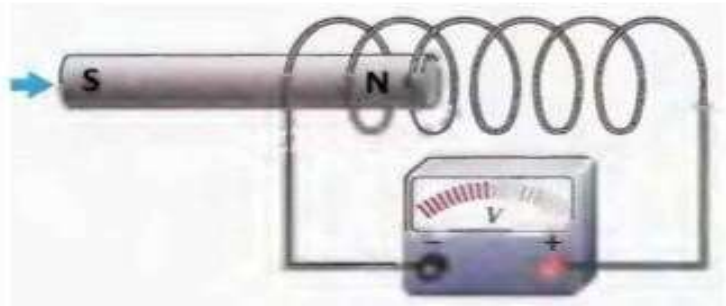
l = Length of conductor passing through the magnetic field (meters)

v = Velocity of conductor (meters per second).

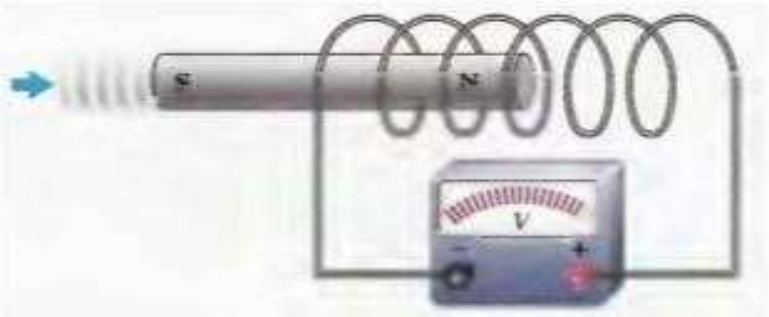


Study:

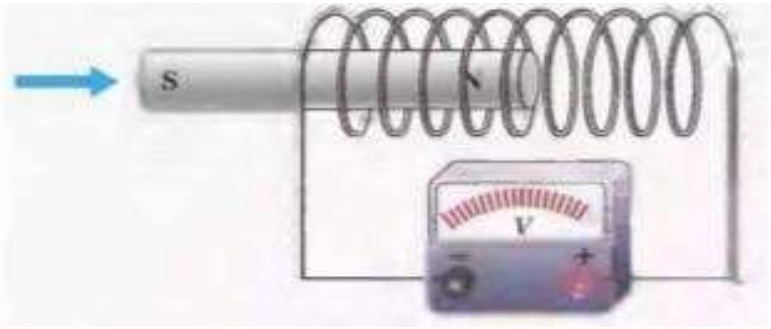
1. Magnet is moved at certain rate and certain voltage is produced.



2. Magnet is moved at faster rate and creating a greater induced voltage.



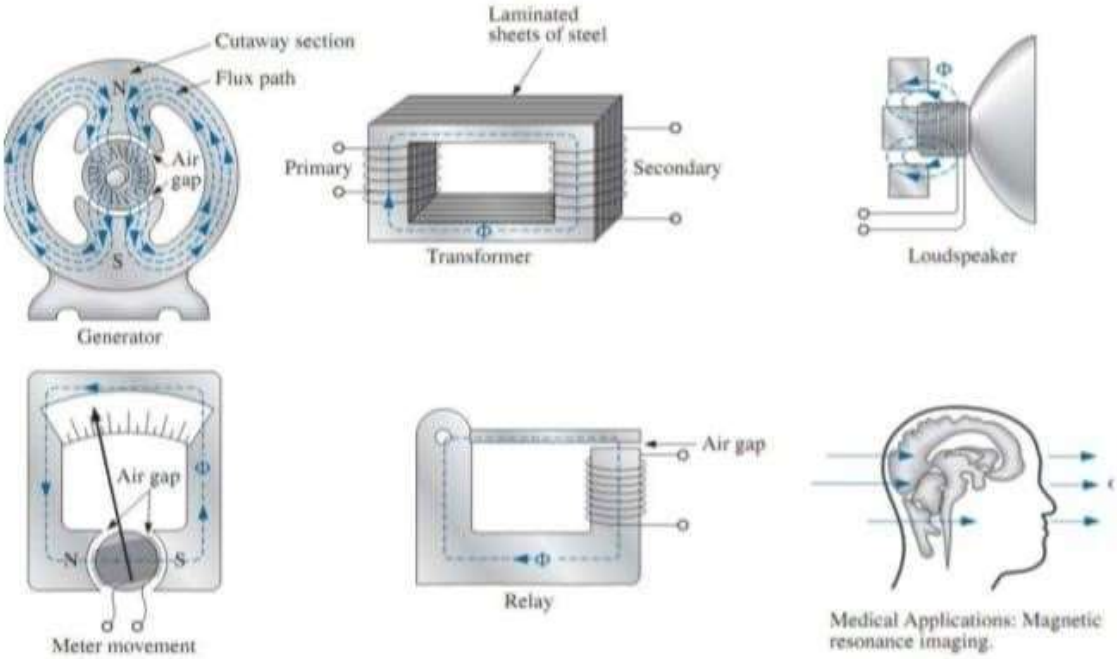
3. Magnet moved at same speed through coil that has greater number of turns and greater voltage is induced.



Some other applications of electromagnetic induction:

The principles of electromagnetic induction are applied in many devices and systems, including:

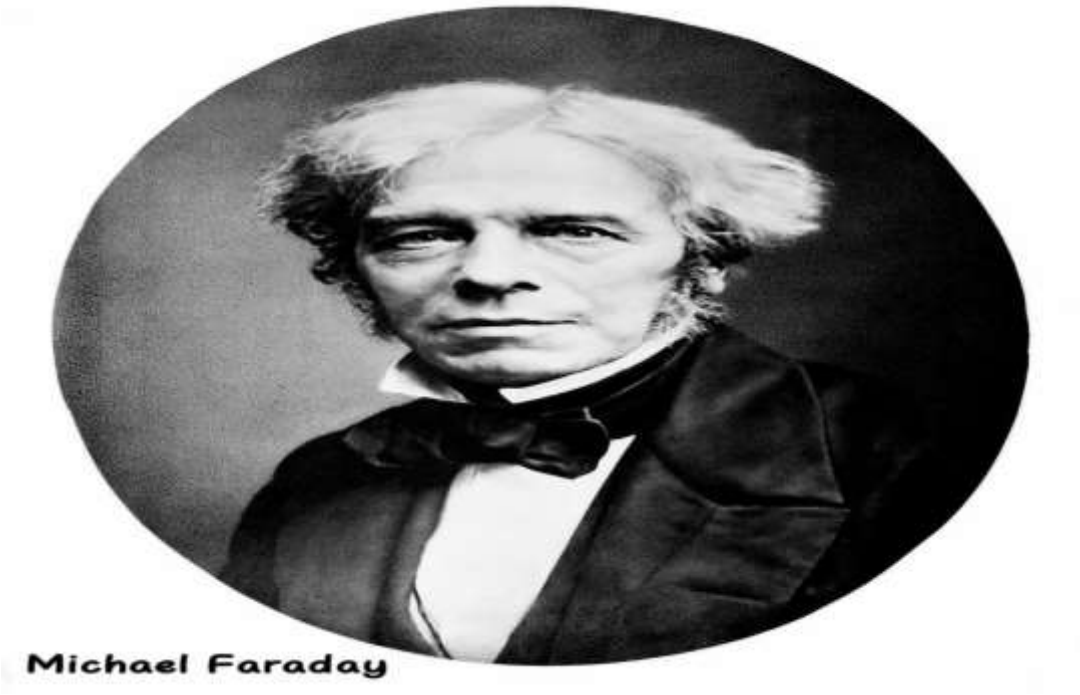
- Electrical generators
  - Induction motor
  - Induction sealing
- Inductive charging
  - Transformers
  - Wireless energy transfer



PRECAUTIONS:

- Keep yourself safe from high voltage.
- Use & handle lab instruments with care.
- Do not scratch insulated copper wire while making loop.
- Do not use digital voltmeter for above demonstration.
- There should not be any electrical energy around the apparatus that can affect the magnetic field of bars or compass.
- The position of the apparatus should not be changed during the experiment. Changing position may affect the final results.

Summary:



Faraday's Law of Electromagnetic Induction, first observed and published by Michael Faraday in the mid-nineteenth century, describes a very important electromagnetic concept. Although its mathematical representations are cryptic, the essence of Faraday's is not hard to grasp: it relates an induced electric potential or voltage to a dynamic magnetic field. This concept has many far reaching ramifications that touch our lives in many ways: from the shining of the sun, to the convenience of mobile communications, to electricity to power our homes. We can all appreciate the profound impact Faraday's Law has on us.

EXPERIMENT

WIRELESS POWER TRANSMISSION

Aim:

The phenomenon of electromagnetic induction is proved by "Wireless Power Transfer" which is one of the applications of EMI.

Materials Required:

- 9V Battery
- Battery cap
- 1 LED (3V any colour)
- 1 Resistance of (any 15k – 20k) ohm
- 1 Transistor (2N2222A)
- Some copper wire (0.4mm – 0.6mm)
- Blade, cardboard, any cylindrical tube, match box & tape.

Procedure:

- Take any cylindrical tube and some copper wire. Make a coil by circulating the copper wire around the cylindrical tube of about (38) thirty eight turns (this is coil 1).
- Make another coil of copper wire by the same above mentioned method of about (19) nineteen turns initially and take out a loop wire and again make (19) nineteen turns (this is coil 2).
- Take the coil 1, remove the copper coating at its two ends carefully with the help of a blade by heating the two ends of the coil.
- Take LED bulb, connect the two pins of bulb with the two ends of coil.
- Take the coil 2, remove the copper coating carefully as done before.
- Take a transistor, connect resistor in its centre pin. Connect one pin of transistor to the one wire coil 2 and connect another wire of coil 2 with the resistor connected transistor pin.
- Take the battery cap, connect its positive wire to the loop wire of coil 2 and connect its negative wire to the remaining last pin of transistor.
- Connect the battery cap with 9V battery.
- With the help of tape connect coil 1 in one cardboard piece and coil 2 in another cardboard piece.

Observation:

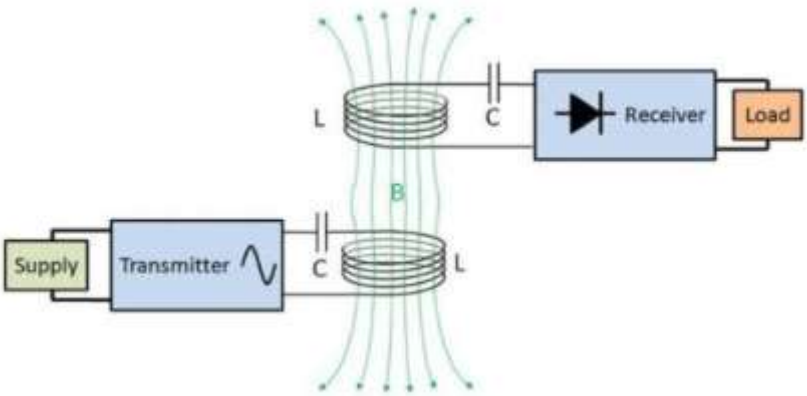
- When coil 1 is brought near to coil 2, the LED bulb glows.
- When coil 1 is taken away from coil 2, the LED bulb does not glow.

Inference:

Inductive coupling is one of the methods that have been used in wireless power transfer system. In this system, it uses two separate coils which is coil 1 as first coil and coil 2 which is the second coil. The coil 2 will attach to the source so it becomes transmitter and the coil 1 will attach with the load therefore it becomes the receiver.

The power will transmit from the coil 2 to coil 1 by electromagnetic wave that was created by the coil 2. This is because when the current goes through conductor (coil) the magnetic flux will be produced. So, the magnetic flux that has been produced from the coil 2, will move to the coil 1 and cut the coil. From basic electromagnetic induction, when the magnetic flux cut the conductor, electric current will induce. After that, the

current will flow to the load as an AC current. As a result in the above experiment the bulb glows.



The block diagram of wireless power transfer induction coupling method.

Precautions:

- ✓ Do all the connections carefully
- ✓ Make sure all the connections are very tight
- ✓ Remove the coating (Insulation) of copper wire carefully
- ✓ The copper wire thickness should be between (0.4mm to 0.6mm)

Result:

The phenomenon of electromagnetic induction is proved.

Image of the above completed ‘Experiment’

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